

**Objective : To create strategy and backtest the following conditions using OOPS concept.**

**Rules : Buy the underlying asset when 10 sma crosses above 20 sma and 20 sma crosses 30 sma , and vice versa for SELL**

**Also optimize the strategy and find out the best asset and combination. Also compare with EMA.**

In [22]: *### importing all the necessary libraries*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
import datetime
import yfinance as yf
import quantstats as qs
```

In [47]:

```
ticker = "SPY"
end = datetime.date(2026,8,30)
start = end - pd.Timedelta(days = 365*10)
sma1 = 50
sma2 = 100
sma3 = 200
```

In [48]: **class** Backtesting3sma:

```
    def __init__(self, ticker, start, end, sma1, sma2, sma3):
        self.ticker = ticker
        self.start = start
        self.end = end
        self.sma1 = sma1
        self.sma2 = sma2
        self.sma3 = sma3
        self.fetch_data()
        self.indicator()
        self.signal()
        self.position()
        self.returns()

    def fetch_data(self):

        self.data = yf.download(self.ticker, self.start, self.end) ## here we downloaded the data from Yfinance
        self.data = self.data.droplevel(1, axis = 1) ## here i am deleting multilevel index from the fetched data, axis1 rej

    def indicator(self):

        self.data["sma1"] = self.data["Close"].rolling(window = self.sma1).mean() ## here we are calculating the rolling mean ;
        self.data["sma2"] = self.data["Close"].rolling(window = self.sma2).mean()
        self.data["sma3"] = self.data["Close"].rolling(window = self.sma3).mean()

    def signal(self): ## here i am defining rules for both Long and short, if Long then 1, if short then -1

        self.data["signal"] = np.where((self.data["sma1"] > self.data["sma2"])&
                                         (self.data["sma2"] > self.data["sma3"])&
                                         (self.data["sma1"].shift(1) < self.data["sma2"].shift(1)) |
                                         (self.data["sma2"].shift(1) < self.data["sma3"].shift(1)), 1, 0)

        self.data["signal"] = np.where((self.data["sma1"] < self.data["sma2"])&
                                         (self.data["sma2"] < self.data["sma3"])&
                                         (self.data["sma1"].shift(1) > self.data["sma2"].shift(1)) |
                                         (self.data["sma2"].shift(1) > self.data["sma3"].shift(1)), -1, self.data["signal"])

    def position(self): ## here i am forward filling the data so that all my zeros are replaced with the signals

        self.data["position"] = self.data["signal"].replace(to_replace=0, method = "ffill")

    def returns(self):

        self.data["bnh"] = np.log(self.data["Close"] / self.data["Close"].shift(1)) ### buy and hold returns.
        self.data["strategy_returns"] = self.data["position"].shift(1) * self.data["bnh"] ## the strategy returns.
        print("The strategy returns:" , np.round(self.data["strategy_returns"].cumsum()[-1], 2))
        return self.data["strategy_returns"].cumsum()[-1]

    def analysis(self):

        self.data[["bnh", "strategy_returns"].cumsum().plot(figsize = (15,9), grid = True)
        self.data[["sma1", "sma2", "sma3", "position"]].plot(figsize = (15,9), secondary_y = "position")
```

```
qs.reports.basic(self.data["strategy_returns"])
```

```
In [80]: d = Backtesting3sma(ticker, start , end , sma1 , sma2 , sma3)
```

```
[*****100%*****] 1 of 1 completed  
The strategy returns: -0.61
```

```
In [81]: d.analysis()
```

## Performance Metrics

| Parameter      | Value |
|----------------|-------|
| Risk-Free Rate | 0.0%  |
| Periods/Year   | 252   |
| Compounded     | Yes   |
| Match Dates    | Yes   |

|                | Strategy   |
|----------------|------------|
| Start Period   | 2016-09-02 |
| End Period     | 2026-08-28 |
| Risk-Free Rate | 0.0%       |
| Time in Market | 92.0%      |

|                   |         |
|-------------------|---------|
| Cumulative Return | -53.82% |
| CAGR %            | -7.46%  |

|                    |        |
|--------------------|--------|
| Sharpe             | -0.34  |
| Prob. Sharpe Ratio | 13.96% |
| Sortino            | -0.5   |
| Sortino/√2         | -0.35  |
| Omega              | 0.93   |

|                     |            |
|---------------------|------------|
| Max Drawdown        | -60.21%    |
| Max DD Date         | 2026-08-13 |
| Max DD Period Start | 2020-03-24 |
| Max DD Period End   | 2026-08-28 |
| Longest DD Days     | 2349       |

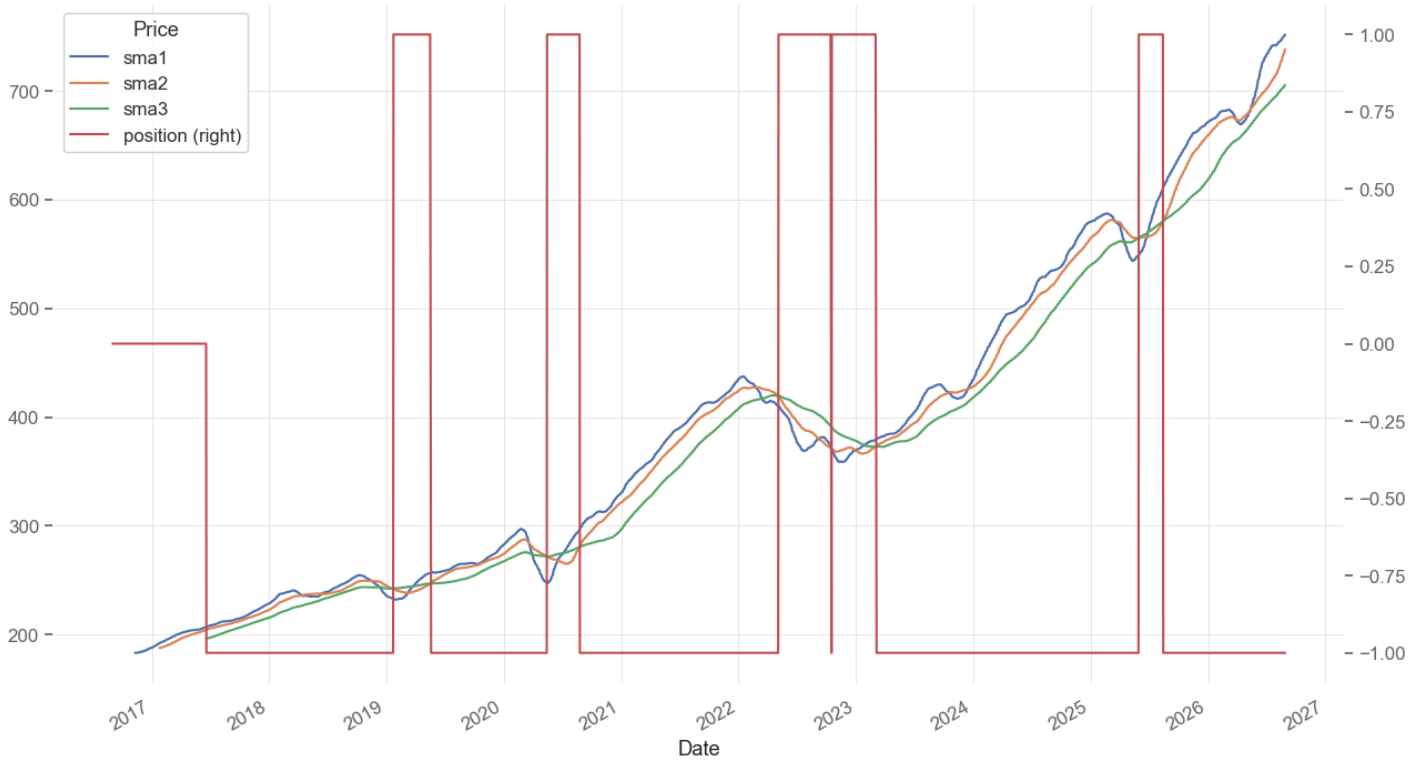
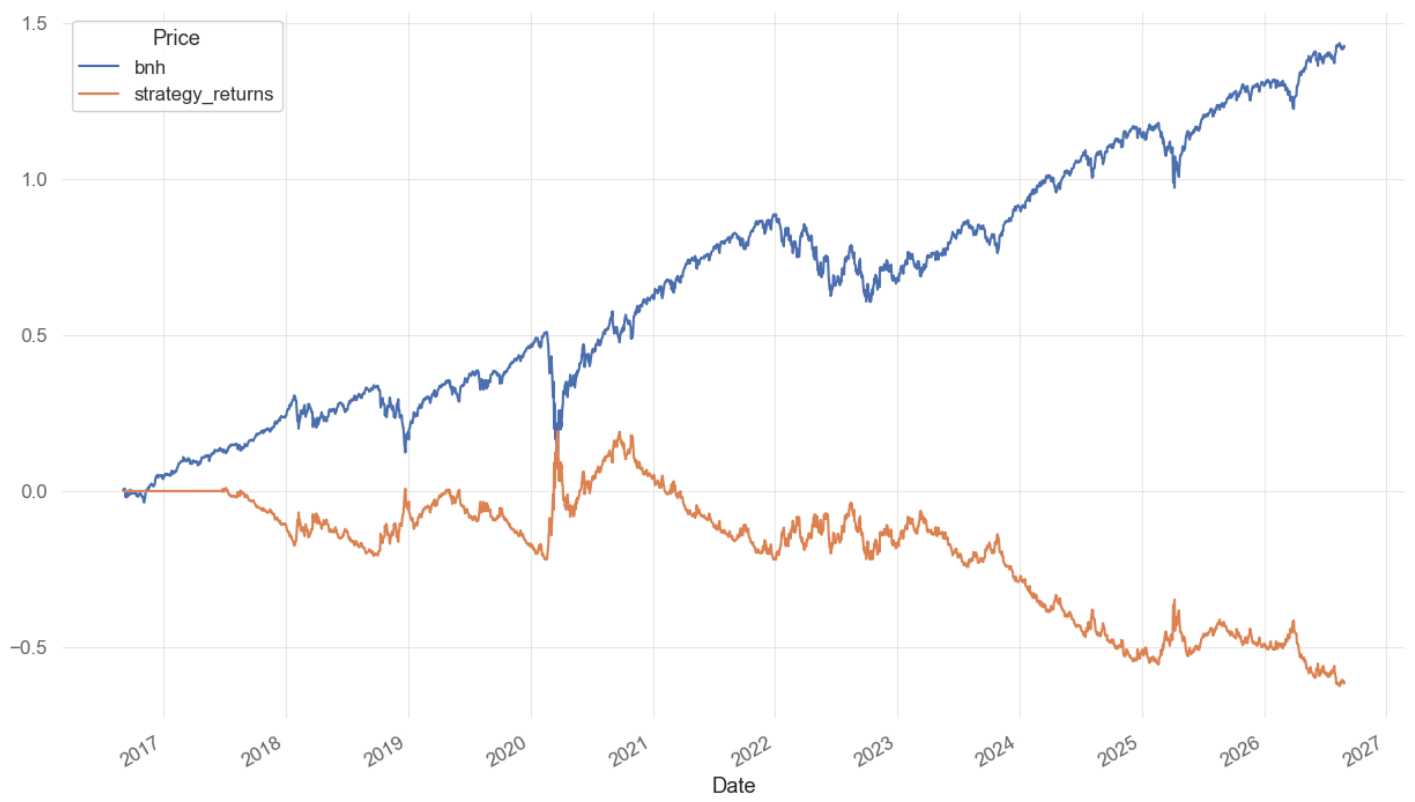
|                 |       |
|-----------------|-------|
| Gain/Pain Ratio | -0.07 |
| Gain/Pain (1M)  | -0.26 |

|                    |      |
|--------------------|------|
| Payoff Ratio       | 1.12 |
| Profit Factor      | 0.93 |
| Common Sense Ratio | 1.03 |
| CPC Index          | 0.47 |
| Tail Ratio         | 1.11 |
| Outlier Win Ratio  | 4.52 |
| Outlier Loss Ratio | 3.88 |

|                 |         |
|-----------------|---------|
| MTD             | -2.94%  |
| 3M              | -2.93%  |
| 6M              | -11.74% |
| YTD             | -12.36% |
| 1Y              | -17.55% |
| 3Y (ann.)       | -15.03% |
| 5Y (ann.)       | -10.57% |
| 10Y (ann.)      | -7.46%  |
| All-time (ann.) | -7.46%  |

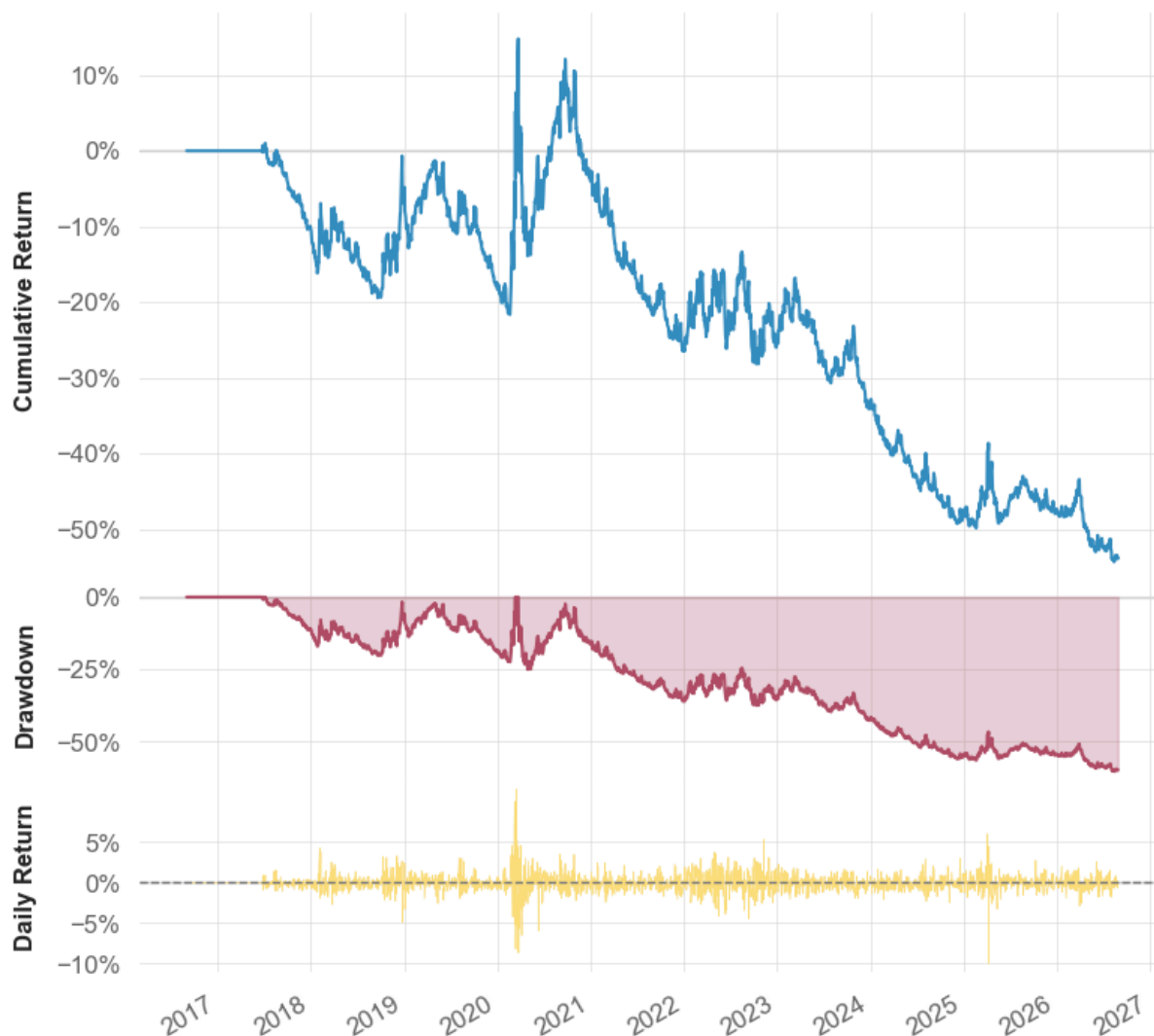
|                    |        |
|--------------------|--------|
| Avg. Drawdown      | -16.2% |
| Avg. Drawdown Days | 557    |
| Recovery Factor    | 1.02   |
| Ulcer Index        | 0.33   |
| Serenity Index     | -0.04  |

## Strategy Visualization



# Portfolio Summary

1 Sep '16 - 28 Aug '26 ; Sharpe: -0.34



Strategy - Monthly Returns (%)

|      | JAN   | FEB   | MAR   | APR    | MAY   | JUN   | JUL   | AUG   | SEP   | OCT   | NOV   | DEC   |
|------|-------|-------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|
| 2016 | 0.00  | 0.00  | 0.00  | 0.00   | 0.00  | 0.00  | 0.00  | 0.00  | 0.00  | 0.00  | 0.00  | 0.00  |
| 2017 | 0.00  | 0.00  | 0.00  | 0.00   | 0.00  | 0.49  | -2.03 | -0.33 | -1.99 | -2.31 | -2.98 | -1.21 |
| 2018 | -5.37 | 3.48  | 2.65  | -0.63  | -2.42 | -0.60 | -3.60 | -3.12 | -0.60 | 7.18  | -1.96 | 9.28  |
| 2019 | -2.91 | 3.21  | 1.75  | 4.07   | -0.48 | -6.56 | -1.52 | 1.48  | -1.94 | -2.23 | -3.51 | -2.85 |
| 2020 | -0.02 | 8.33  | 10.45 | -11.87 | 7.65  | 1.41  | 5.81  | 3.79  | 3.64  | 2.38  | -9.92 | -3.61 |
| 2021 | 0.93  | -2.78 | -4.46 | -5.07  | -0.73 | -2.22 | -2.43 | -2.92 | 4.81  | -6.61 | 0.74  | -4.54 |
| 2022 | 5.42  | 2.85  | -3.85 | 9.34   | -1.04 | -8.59 | 9.04  | -4.24 | -9.48 | 5.30  | 5.24  | -5.89 |
| 2023 | 6.17  | -2.60 | 0.31  | -1.62  | -0.53 | -6.14 | -3.20 | 1.58  | 4.93  | 2.13  | -8.43 | -4.40 |
| 2024 | -1.62 | -5.02 | -3.20 | 4.12   | -4.86 | -3.42 | -1.28 | -2.43 | -2.13 | 0.85  | -5.69 | 2.38  |
| 2025 | -2.69 | 1.22  | 5.71  | -0.22  | -6.03 | 5.09  | 2.28  | 1.96  | -3.46 | -2.41 | -0.28 | -0.11 |
| 2026 | -1.49 | 0.81  | 5.05  | -9.58  | -5.04 | 0.91  | -0.10 | -2.94 | 0.00  | 0.00  | 0.00  | 0.00  |

```
In [54]: ## to know what combinations yields me the most so we do optimize the above strategy
ma1 = [7,11,15,30]
ma2 = [50,60,75,100]
ma3 = [125,130,150,200] ## here we can add multiple combination of our choice and see the best outcome

SMA1 = []
SMA2 = []
SMA3 = []
outcome = []

for i in ma1:
    for j in ma2:
        for k in ma3:
            print("for" ,i , j , k)
```

```
a= Backtesting3sma(ticker , start , end , i , j , k)
```

```
SMA1.append(i)
SMA2.append(j)
SMA3.append(k)
outcome.append(a.returns())
```

```
results = pd.DataFrame({"SMA1" : SMA1 , "SMA2" : SMA2 , "SMA3" : SMA3 , "outcome" : outcome})
results.sort_values(by="outcome" , ascending=False)
```

```
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```

```
for 7 50 125
```

```
The strategy returns: -0.35
```

```
The strategy returns: -0.35
```

```
for 7 50 130
```

```
The strategy returns: -0.22
```

```
The strategy returns: -0.22
```

```
for 7 50 150
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.73
```

```
The strategy returns: -0.73
```

```
for 7 50 200
```

```
The strategy returns: -0.58
```

```
The strategy returns: -0.58
```

```
for 7 60 125
```

```
The strategy returns: -0.38
```

```
The strategy returns: -0.38
```

```
for 7 60 130
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.18
```

```
The strategy returns: -0.18
```

```
for 7 60 150
```

```
The strategy returns: -0.45
```

```
The strategy returns: -0.45
```

```
for 7 60 200
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.48
```

```
The strategy returns: -0.48
```

```
for 7 75 125
```

```
The strategy returns: -0.21
```

```
The strategy returns: -0.21
```

```
for 7 75 130
```

```
The strategy returns: -0.57
```

```
The strategy returns: -0.57
```

```
for 7 75 150
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.55
```

```
The strategy returns: -0.55
```

```
for 7 75 200
```

```
The strategy returns: -0.58
```

```
The strategy returns: -0.58
```

```
for 7 100 125
```

```
The strategy returns: -0.29
```

```
The strategy returns: -0.29
```

```
for 7 100 130
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.35
```

```
The strategy returns: -0.35
```

```
for 7 100 150
```

```
The strategy returns: -0.41
```

```
The strategy returns: -0.41
```

```
for 7 100 200
```

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.55
```

```
The strategy returns: -0.55
```

```
for 11 50 125
```

```
The strategy returns: -0.33
```

```
The strategy returns: -0.33
```

```
for 11 50 130
```

```
The strategy returns: -0.2
```

```
The strategy returns: -0.2
```

```
for 11 50 150
```

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.7  
The strategy returns: -0.7  
for 11 50 200  
The strategy returns: -0.55  
The strategy returns: -0.55  
for 11 60 125

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.31  
The strategy returns: -0.31  
for 11 60 130  
The strategy returns: -0.11  
The strategy returns: -0.11  
for 11 60 150  
The strategy returns: -0.38  
The strategy returns: -0.38  
for 11 60 200

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.42  
The strategy returns: -0.42  
for 11 75 125  
The strategy returns: -0.12  
The strategy returns: -0.12  
for 11 75 130  
The strategy returns: -0.48  
The strategy returns: -0.48  
for 11 75 150

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.46  
The strategy returns: -0.46  
for 11 75 200  
The strategy returns: -0.46  
The strategy returns: -0.46  
for 11 100 125  
The strategy returns: -0.3  
The strategy returns: -0.3  
for 11 100 130

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.35  
The strategy returns: -0.35  
for 11 100 150  
The strategy returns: -0.43  
The strategy returns: -0.43  
for 11 100 200  
The strategy returns: -0.6  
The strategy returns: -0.6  
for 15 50 125

```
[*****100%*****] 1 of 1 completed
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[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.31  
The strategy returns: -0.31  
for 15 50 130  
The strategy returns: -0.18  
The strategy returns: -0.18  
for 15 50 150  
The strategy returns: -0.68

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.68  
for 15 50 200  
The strategy returns: -0.53  
The strategy returns: -0.53  
for 15 60 125  
The strategy returns: -0.41  
The strategy returns: -0.41  
for 15 60 130

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.21  
The strategy returns: -0.21  
for 15 60 150  
The strategy returns: -0.48  
The strategy returns: -0.48  
for 15 60 200  
The strategy returns: -0.51  
The strategy returns: -0.51  
for 15 75 125

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The strategy returns: -0.1  
The strategy returns: -0.1  
for 15 75 130  
The strategy returns: -0.46  
The strategy returns: -0.46  
for 15 75 150

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The strategy returns: -0.44  
The strategy returns: -0.44  
for 15 75 200  
The strategy returns: -0.44  
The strategy returns: -0.44  
for 15 100 125  
The strategy returns: -0.33  
The strategy returns: -0.33  
for 15 100 130

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The strategy returns: -0.35  
The strategy returns: -0.35  
for 15 100 150  
The strategy returns: -0.44  
The strategy returns: -0.44  
for 15 100 200  
The strategy returns: -0.61  
The strategy returns: -0.61  
for 30 50 125

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
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The strategy returns: -0.21  
The strategy returns: -0.21  
for 30 50 130  
The strategy returns: -0.07  
The strategy returns: -0.07  
for 30 50 150

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The strategy returns: -0.58  
The strategy returns: -0.58  
for 30 50 200  
The strategy returns: -0.43  
The strategy returns: -0.43  
for 30 60 125  
The strategy returns: -0.42  
The strategy returns: -0.42  
for 30 60 130

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The strategy returns: -0.21  
The strategy returns: -0.21  
for 30 60 150  
The strategy returns: -0.45  
The strategy returns: -0.45  
for 30 60 200

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The strategy returns: -0.52  
The strategy returns: -0.52  
for 30 75 125  
The strategy returns: -0.12  
The strategy returns: -0.12  
for 30 75 130  
The strategy returns: -0.48  
The strategy returns: -0.48  
for 30 75 150

```
[*****100%*****] 1 of 1 completed
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```

```
The strategy returns: -0.46
The strategy returns: -0.46
for 30 75 200
The strategy returns: -0.46
The strategy returns: -0.46
for 30 100 125
The strategy returns: -0.28
The strategy returns: -0.28
for 30 100 130
```

```
[*****100%*****] 1 of 1 completed
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[*****100%*****] 1 of 1 completed
```

```
The strategy returns: -0.33
The strategy returns: -0.33
for 30 100 150
The strategy returns: -0.39
The strategy returns: -0.39
for 30 100 200
The strategy returns: -0.56
The strategy returns: -0.56
```

```
Out[54]:
```

|     | SMA1 | SMA2 | SMA3 | outcome   |
|-----|------|------|------|-----------|
| 49  | 30   | 50   | 130  | -0.073659 |
| 40  | 15   | 75   | 125  | -0.101233 |
| 21  | 11   | 60   | 130  | -0.109978 |
| 56  | 30   | 75   | 125  | -0.118007 |
| 24  | 11   | 75   | 125  | -0.122045 |
| ... | ...  | ...  | ...  | ...       |
| 31  | 11   | 100  | 200  | -0.595655 |
| 47  | 15   | 100  | 200  | -0.610892 |
| 34  | 15   | 50   | 150  | -0.683126 |
| 18  | 11   | 50   | 150  | -0.703020 |
| 2   | 7    | 50   | 150  | -0.726193 |

64 rows × 4 columns

```
In [52]: assets = [ "AAPL",
                    "TSLA",
                    "RELIANCE.NS",
                    "INFY.NS",
                    "SPY",
                    "BTC-USD",
                    "ETH-USD",
                    "GC=F",
                    "CL=F",
                    "EURUSD=X"]
asset1 = []
out_returns = []
for x in assets:
    print("The asset is:", x)
    a = Backtesting3sma(x, start ,end , sma1 , sma1, sma3)
    asset1.append(x)
    out_returns.append(a.returns())

Results = pd.DataFrame({"assets1" : asset1 , "out_returns" : out_returns})
Results
```

```
[*****100%*****] 1 of 1 completed
```

The asset is: AAPL

```
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[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.18  
The strategy returns: -0.18  
The asset is: TSLA  
The strategy returns: -0.57  
The strategy returns: -0.57  
The asset is: RELIANCE.NS  
The strategy returns: -0.07  
The strategy returns: -0.07  
The asset is: INFY.NS  
The strategy returns: 0.44  
The strategy returns: 0.44  
The asset is: SPY

```
[*****100%*****] 1 of 1 completed
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```

The strategy returns: -0.48  
The strategy returns: -0.48  
The asset is: BTC-USD  
The strategy returns: -3.59  
The strategy returns: -3.59  
The asset is: ETH-USD  
The strategy returns: -1.98  
The strategy returns: -1.98  
The asset is: GC=F

```
[*****100%*****] 1 of 1 completed
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[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.54  
The strategy returns: -0.54  
The asset is: CL=F  
The strategy returns: 1.46  
The strategy returns: 1.46  
The asset is: EURUSD=X  
The strategy returns: 0.02  
The strategy returns: 0.02

```
Out[52]:
```

|   | assets1     | out_returns |
|---|-------------|-------------|
| 0 | AAPL        | -0.175360   |
| 1 | TSLA        | -0.571657   |
| 2 | RELIANCE.NS | -0.070240   |
| 3 | INFY.NS     | 0.439101    |
| 4 | SPY         | -0.482020   |
| 5 | BTC-USD     | -3.592197   |
| 6 | ETH-USD     | -1.980166   |
| 7 | GC=F        | -0.540276   |
| 8 | CL=F        | 1.455761    |
| 9 | EURUSD=X    | 0.024570    |

```
In [64]: assets = [ "AAPL",
                    "TSLA",
                    "RELIANCE.NS",
                    "INFY.NS",
                    "SPY",
                    "BTC-USD",
                    "ETH-USD",
                    "GC=F",
                    "CL=F",
                    "EURUSD=X"]
ma1 = [15,30]
ma2 = [75,100]
ma3 = [150,200]

Stock_list = []
MA1 = []
MA2 = []
MA3 = []
ROI = []

for a in assets:
    for b in ma1:
        for c in ma2:
            for d in ma3:
                print("The stock is :", a)

                w = Backtesting3sma(a , start , end , b , c, d)
                Stock_list.append(a)
                MA1.append(b)
```

```
MA2.append(c)
MA3.append(d)
ROI.append(w.returns())
```

The stock is : AAPL

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The strategy returns: -0.01

The strategy returns: -0.01

The stock is : AAPL

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The strategy returns: -0.19

The strategy returns: -0.19

The stock is : AAPL

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The strategy returns: -0.54

The strategy returns: -0.54

The stock is : AAPL

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The strategy returns: -0.67

The strategy returns: -0.67

The stock is : AAPL

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The strategy returns: -0.1

The strategy returns: -0.1

The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.23

The strategy returns: -0.23

The stock is : AAPL

The strategy returns: -0.48

The strategy returns: -0.48

The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.59

The strategy returns: -0.59

The stock is : TSLA

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The strategy returns: -3.5

The strategy returns: -3.5

The stock is : TSLA

The strategy returns: -0.84

The strategy returns: -0.84

The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -2.08

The strategy returns: -2.08

The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.6

The strategy returns: -0.6

The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -3.04

The strategy returns: -3.04

The stock is : TSLA

The strategy returns: -0.42

The strategy returns: -0.42

The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -2.16

The strategy returns: -2.16

The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.87

The strategy returns: -0.87

The stock is : RELIANCE.NS

The strategy returns: 1.51

The strategy returns: 1.51

The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.16

The strategy returns: 0.16

The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.89  
The strategy returns: 0.89  
The stock is : RELIANCE.NS  
The strategy returns: -0.13  
The strategy returns: -0.13  
The stock is : RELIANCE.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 1.57  
The strategy returns: 1.57  
The stock is : RELIANCE.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 0.25  
The strategy returns: 0.25  
The stock is : RELIANCE.NS  
The strategy returns: 0.94  
The strategy returns: 0.94  
The stock is : RELIANCE.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.1  
The strategy returns: -0.1  
The stock is : INFY.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 0.29  
The strategy returns: 0.29  
The stock is : INFY.NS  
The strategy returns: 0.79  
The strategy returns: 0.79  
The stock is : INFY.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 0.41  
The strategy returns: 0.41  
The stock is : INFY.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.05  
The strategy returns: -0.05  
The stock is : INFY.NS  
The strategy returns: 0.08  
The strategy returns: 0.08  
The stock is : INFY.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 0.77  
The strategy returns: 0.77  
The stock is : INFY.NS  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: 0.52  
The strategy returns: 0.52  
The stock is : INFY.NS  
The strategy returns: 0.1  
The strategy returns: 0.1  
The stock is : SPY  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.44  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.44  
The stock is : SPY  
The strategy returns: -0.44  
The strategy returns: -0.44  
The stock is : SPY  
The strategy returns: -0.44  
The strategy returns: -0.44  
The stock is : SPY  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.61  
The strategy returns: -0.61  
The stock is : SPY  
The strategy returns: -0.46  
The strategy returns: -0.46  
The stock is : SPY  
The strategy returns: -0.46  
The strategy returns: -0.46  
The stock is : SPY  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.39  
The strategy returns: -0.39  
The stock is : SPY  
The strategy returns: -0.56  
The strategy returns: -0.56  
The stock is : BTC-USD  
The strategy returns: -4.37  
The strategy returns: -4.37  
The stock is : BTC-USD

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -1.99  
The strategy returns: -1.99  
The stock is : BTC-USD  
The strategy returns: -4.24  
The strategy returns: -4.24  
The stock is : BTC-USD  
The strategy returns: -3.2  
The strategy returns: -3.2  
The stock is : BTC-USD

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -4.55  
The strategy returns: -4.55  
The stock is : BTC-USD  
The strategy returns: -2.26  
The strategy returns: -2.26  
The stock is : BTC-USD  
The strategy returns: -4.37  
The strategy returns: -4.37  
The stock is : BTC-USD

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -3.06  
The strategy returns: -3.06  
The stock is : ETH-USD  
The strategy returns: -1.41  
The strategy returns: -1.41  
The stock is : ETH-USD  
The strategy returns: -2.19  
The strategy returns: -2.19  
The stock is : ETH-USD

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -2.63  
The strategy returns: -2.63  
The stock is : ETH-USD  
The strategy returns: -2.58  
The strategy returns: -2.58  
The stock is : ETH-USD  
The strategy returns: -1.85  
The strategy returns: -1.85  
The stock is : ETH-USD

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -2.66  
The strategy returns: -2.66  
The stock is : ETH-USD  
The strategy returns: -2.72  
The strategy returns: -2.72  
The stock is : ETH-USD  
The strategy returns: -2.64  
The strategy returns: -2.64  
The stock is : GC=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.69  
The strategy returns: -0.69  
The stock is : GC=F  
The strategy returns: -0.44  
The strategy returns: -0.44  
The stock is : GC=F  
The strategy returns: -0.8  
The strategy returns: -0.8  
The stock is : GC=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.59  
The strategy returns: -0.59  
The stock is : GC=F  
The strategy returns: -0.69  
The strategy returns: -0.69  
The stock is : GC=F  
The strategy returns: -0.42  
The strategy returns: -0.42  
The stock is : GC=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.71  
The strategy returns: -0.71  
The stock is : GC=F  
The strategy returns: -0.53  
The strategy returns: -0.53  
The stock is : CL=F  
The strategy returns: 1.78  
The strategy returns: 1.78  
The stock is : CL=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 1.9  
The strategy returns: 1.9  
The stock is : CL=F  
The strategy returns: 2.24  
The strategy returns: 2.24  
The stock is : CL=F  
The strategy returns: 0.5  
The strategy returns: 0.5  
The stock is : CL=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 1.98  
The strategy returns: 1.98  
The stock is : CL=F  
The strategy returns: 2.14  
The strategy returns: 2.14  
The stock is : CL=F  
The strategy returns: 2.26  
The strategy returns: 2.26  
The stock is : CL=F

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.46  
The strategy returns: 0.46  
The stock is : EURUSD=X  
The strategy returns: 0.04  
The strategy returns: 0.04  
The stock is : EURUSD=X  
The strategy returns: -0.02  
The strategy returns: -0.02  
The stock is : EURUSD=X

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.03  
The strategy returns: 0.03  
The stock is : EURUSD=X  
The strategy returns: -0.03  
The strategy returns: -0.03  
The stock is : EURUSD=X  
The strategy returns: 0.06  
The strategy returns: 0.06  
The stock is : EURUSD=X

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.04  
The strategy returns: 0.04  
The stock is : EURUSD=X  
The strategy returns: 0.0  
The strategy returns: 0.0  
The stock is : EURUSD=X  
The strategy returns: -0.04  
The strategy returns: -0.04

```
In [66]: multiple_results = pd.DataFrame({"Stock_list" : Stock_list , "MA1" : MA1 , "MA2" : MA2 , "MA3" : MA3, "ROI" : ROI})
multiple_results.sort_values(by="ROI" ,ascending=False)
```

Out[66]:

|     | Stock_list | MA1 | MA2 | MA3 | ROI       |
|-----|------------|-----|-----|-----|-----------|
| 70  | CL=F       | 30  | 100 | 150 | 2.261271  |
| 66  | CL=F       | 15  | 100 | 150 | 2.242674  |
| 69  | CL=F       | 30  | 75  | 200 | 2.144641  |
| 68  | CL=F       | 30  | 75  | 150 | 1.977497  |
| 65  | CL=F       | 15  | 75  | 200 | 1.902250  |
| ... | ...        | ... | ... | ... | ...       |
| 8   | TSLA       | 15  | 75  | 150 | -3.501250 |
| 42  | BTC-USD    | 15  | 100 | 150 | -4.236977 |
| 46  | BTC-USD    | 30  | 100 | 150 | -4.367782 |
| 40  | BTC-USD    | 15  | 75  | 150 | -4.370102 |
| 44  | BTC-USD    | 30  | 75  | 150 | -4.546782 |

80 rows × 5 columns

```
In [ ]:
```

```
In [67]: ## now we were performing using SMA's Lets switch to EMA's
```

```
In [74]: class BacktestingEma(Backtesting3sma): ## This is Inhereitence method , passing all the attributes and methods to child class

    def indicator(self):
        self.data["sma1"] = self.data["Close"].ewm(span = self.sma1).mean()
        self.data["sma2"] = self.data["Close"].ewm(span = self.sma2).mean()
        self.data["sma3"] = self.data["Close"].ewm(span = self.sma3).mean()
```

```
In [75]: q = BacktestingEma(ticker, start , end ,sma1 ,sma2 , sma3)
```

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
The strategy returns: -0.65

```
In [78]: assets = [ "AAPL",
    "TSLA",
    "RELIANCE.NS",
    "INFY.NS",
    "SPY",
    "BTC-USD",
    "ETH-USD",
    "GC=F",
    "CL=F",
    "EURUSD=X"]
ma1 = [15,30]
ma2 = [75,100]
ma3 = [150,200]

Stock_list = []
MA1 = []
MA2 = []
MA3 = []
ROI = []

for a in assets:
    for b in ma1:
        for c in ma2:
            for d in ma3:
                print("The stock is :" , a)

                w = BacktestingEma(a , start , end , b , c, d)
                Stock_list.append(a)
                MA1.append(b)
                MA2.append(c)
                MA3.append(d)
                ROI.append(w.returns())
```

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed  
[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The stock is : AAPL  
The strategy returns: -0.87  
The strategy returns: -0.87  
The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.78  
The strategy returns: -0.78  
The stock is : AAPL  
The strategy returns: -0.53  
The strategy returns: -0.53  
The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.65  
The strategy returns: -0.65  
The stock is : AAPL  
The strategy returns: -0.78  
The strategy returns: -0.78  
The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.69  
The strategy returns: -0.69  
The stock is : AAPL  
The strategy returns: -0.52  
The strategy returns: -0.52  
The stock is : AAPL

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.64  
The strategy returns: -0.64  
The stock is : TSLA  
The strategy returns: 0.71  
The strategy returns: 0.71  
The stock is : TSLA  
The strategy returns: -0.59  
The strategy returns: -0.59  
The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -1.77  
The strategy returns: -1.77  
The stock is : TSLA  
The strategy returns: -0.84  
The strategy returns: -0.84  
The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.89  
The strategy returns: 0.89  
The stock is : TSLA  
The strategy returns: -0.41  
The strategy returns: -0.41  
The stock is : TSLA  
The strategy returns: -1.53  
The strategy returns: -1.53  
The stock is : TSLA

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.68  
The strategy returns: -0.68  
The stock is : RELIANCE.NS  
The strategy returns: -0.45  
The strategy returns: -0.45  
The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.26  
The strategy returns: -0.26  
The stock is : RELIANCE.NS  
The strategy returns: -0.49  
The strategy returns: -0.49  
The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.38  
The strategy returns: -0.38  
The stock is : RELIANCE.NS  
The strategy returns: -0.46  
The strategy returns: -0.46  
The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.28  
The strategy returns: -0.28  
The stock is : RELIANCE.NS  
The strategy returns: -0.5  
The strategy returns: -0.5  
The stock is : RELIANCE.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.4  
The strategy returns: -0.4  
The stock is : INFY.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.65  
The strategy returns: 0.65  
The stock is : INFY.NS  
The strategy returns: 0.48  
The strategy returns: 0.48  
The stock is : INFY.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.18  
The strategy returns: 0.18  
The stock is : INFY.NS  
The strategy returns: 0.23  
The strategy returns: 0.23  
The stock is : INFY.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.57  
The strategy returns: 0.57  
The stock is : INFY.NS  
The strategy returns: 0.35  
The strategy returns: 0.35  
The stock is : INFY.NS

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: 0.26  
The strategy returns: 0.26  
The stock is : INFY.NS  
The strategy returns: 0.3  
The strategy returns: 0.3  
The stock is : SPY

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.49  
The strategy returns: -0.49  
The stock is : SPY  
The strategy returns: -0.79  
The strategy returns: -0.79  
The stock is : SPY

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.77  
The strategy returns: -0.77  
The stock is : SPY  
The strategy returns: -0.67  
The strategy returns: -0.67  
The stock is : SPY

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.51  
The strategy returns: -0.51  
The stock is : SPY  
The strategy returns: -0.78  
The strategy returns: -0.78  
The stock is : SPY

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

[\*\*\*\*\*100%\*\*\*\*\*] 1 of 1 completed

The strategy returns: -0.76  
The strategy returns: -0.76  
The stock is : SPY  
The strategy returns: -0.67  
The strategy returns: -0.67  
The stock is : BTC-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -3.76  
The strategy returns: -3.76  
The stock is : BTC-USD  
The strategy returns: -3.56  
The strategy returns: -3.56  
The stock is : BTC-USD  
The strategy returns: -3.92  
The strategy returns: -3.92  
The stock is : BTC-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -3.96  
The strategy returns: -3.96  
The stock is : BTC-USD  
The strategy returns: -3.67  
The strategy returns: -3.67  
The stock is : BTC-USD  
The strategy returns: -3.43  
The strategy returns: -3.43  
The stock is : BTC-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -3.88  
The strategy returns: -3.88  
The stock is : BTC-USD  
The strategy returns: -3.92  
The strategy returns: -3.92  
The stock is : ETH-USD  
The strategy returns: -1.87  
The strategy returns: -1.87  
The stock is : ETH-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -2.1  
The strategy returns: -2.1  
The stock is : ETH-USD  
The strategy returns: -1.78  
The strategy returns: -1.78  
The stock is : ETH-USD  
The strategy returns: -0.97  
The strategy returns: -0.97  
The stock is : ETH-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.88  
The strategy returns: -0.88  
The stock is : ETH-USD  
The strategy returns: -1.11  
The strategy returns: -1.11  
The stock is : ETH-USD  
The strategy returns: -1.55  
The strategy returns: -1.55  
The stock is : ETH-USD

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.74  
The strategy returns: -0.74  
The stock is : GC=F  
The strategy returns: -0.5  
The strategy returns: -0.5  
The stock is : GC=F  
The strategy returns: -0.61  
The strategy returns: -0.61  
The stock is : GC=F

```
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
```

The strategy returns: -0.37  
The strategy returns: -0.37  
The stock is : GC=F  
The strategy returns: -0.56  
The strategy returns: -0.56  
The stock is : GC=F  
The strategy returns: -0.44  
The strategy returns: -0.44  
The stock is : GC=F

```

[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: -0.54
The strategy returns: -0.54
The stock is : GC=F
The strategy returns: -0.33
The strategy returns: -0.33
The stock is : GC=F
The strategy returns: -0.51
The strategy returns: -0.51
The stock is : CL=F
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: 1.27
The strategy returns: 1.27
The stock is : CL=F
The strategy returns: 2.14
The strategy returns: 2.14
The stock is : CL=F
The strategy returns: 1.74
The strategy returns: 1.74
The stock is : CL=F
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: 1.23
The strategy returns: 1.23
The stock is : CL=F
The strategy returns: 1.01
The strategy returns: 1.01
The stock is : CL=F
The strategy returns: 1.95
The strategy returns: 1.95
The stock is : CL=F
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: 1.45
The strategy returns: 1.45
The stock is : CL=F
The strategy returns: 0.93
The strategy returns: 0.93
The stock is : EURUSD=X
The strategy returns: -0.15
The strategy returns: -0.15
The stock is : EURUSD=X
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: 0.01
The strategy returns: 0.01
The stock is : EURUSD=X
The strategy returns: 0.03
The strategy returns: 0.03
The stock is : EURUSD=X
The strategy returns: 0.05
The strategy returns: 0.05
The stock is : EURUSD=X
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed
The strategy returns: -0.14
The strategy returns: -0.14
The stock is : EURUSD=X
The strategy returns: 0.02
The strategy returns: 0.02
The stock is : EURUSD=X
The strategy returns: 0.03
The strategy returns: 0.03
The stock is : EURUSD=X
The strategy returns: 0.06
The strategy returns: 0.06

```

```

In [79]: multiple_results = pd.DataFrame({"Stock_list" : Stock_list , "MA1" : MA1 , "MA2" : MA2 , "MA3" : MA3, "ROI" : ROI})
multiple_results.sort_values(by="ROI" ,ascending=False)

```

Out[79]:

|     | Stock_list | MA1 | MA2 | MA3 | ROI       |
|-----|------------|-----|-----|-----|-----------|
| 65  | CL=F       | 15  | 75  | 200 | 2.141598  |
| 69  | CL=F       | 30  | 75  | 200 | 1.949341  |
| 66  | CL=F       | 15  | 100 | 150 | 1.735246  |
| 70  | CL=F       | 30  | 100 | 150 | 1.446826  |
| 64  | CL=F       | 15  | 75  | 150 | 1.273578  |
| ... | ...        | ... | ... | ... | ...       |
| 40  | BTC-USD    | 15  | 75  | 150 | -3.759662 |
| 46  | BTC-USD    | 30  | 100 | 150 | -3.877458 |
| 47  | BTC-USD    | 30  | 100 | 200 | -3.915356 |
| 42  | BTC-USD    | 15  | 100 | 150 | -3.922848 |
| 43  | BTC-USD    | 15  | 100 | 200 | -3.960746 |

80 rows × 5 columns

In [ ]: